Research Project – (Monetary Theory and Policy)

Financial Deepening and Monetary Transmission: Why the Asset Price Channel Varies Between the US and Pakistan

Researcher:
Muhammad Ghazi

Supervisor:
Dr Mirza Aqeel Baig

Department of Economics
College of Economics and Social Development
Institute of Business Management (IoBM)
Karachi

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ABSTRACT

This study provides a comparative empirical assessment of the Asset Price Monetary Transmission Mechanism in the United States and Pakistan, bridging a critical gap in recent literature by analyzing the post-2022 global tightening cycle. The study utilizes a Vector Error Correction Model (VECM) on monthly data starting from July 2015 to January 2025, the research explores how policy rate innovations (6MKIBOR and Federal Funds Rate) transmit to the real economy through equity and real estate valuations.

Using the Johansen Cointegration findings, this research confirms the presence of a stable long-run relationship among the variables in both economies, the transmission is characterized by near-perfect efficiency; Impulse Response Functions (IRFs) show an immediate and sharp contraction in equity valuations (Dow Jones) following a policy shock, with a smooth, persistent decline in the House Rent Index (HRI). Conversely, the Pakistan model depicts significant structural frictions, where the equity channel (PSX) moves with a gradual and lagged adjustment, and the housing channel is notably "sticky," with a speed of adjustment (ECT) of only 4% per month.

The results highlight contrast in the applicability of the Wealth Effect and Tobin's q mechanisms between developed and emerging markets. While the developed economic such as US has an asset price channel that is responsive whereas in contrary developing nation as Pakistan contains weaker financial system where transmission is sticky and time taking, leaving the economy more reliant on bank-credit channels. These findings suggest that for emerging economies like Pakistan, the effectiveness of monetary policy remains deeply contingent on financial sector development and the formalization of asset markets.

Chapter 1

Introduction

The transmission mechanism of monetary policy is a cornerstone of macroeconomic theory which describe the process through which Central Bank’s actions related to policy fluctuation influence aggregate demand and price stability. While there are other channels such as the interest rate, exchange rate, and credit channels, the Asset Price Channel in particular has gained significant prominence in the modern financial era. This channel operates on the premise that changes in policy rates (such as the Federal Funds Rate or the KIBOR) influence the valuation of financial and real assets, which in turn impacts investment through Tobin’s q theory and household consumption through the wealth effect.

1.1 Background of the Study

In developed economies like the United States, the asset price channel is sophisticated and integrated into their financial system. There are evidences that suggests that US financial markets react almost instantaneously to monetary policy surprises, with profound impacts on both equity valuations and the housing market. However, the effectiveness of this channel in emerging economies, in our case Pakistan, remains a subject of intense debate. In Pakistan, the transmission mechanism is often described as credit-driven, characterized by structural frictions such as a large informal economy, limited financial market depth, and an underdeveloped mortgage market.

1.2 Problem Statement

Despite the theoretical recognition of the asset price channel, there is a significant empirical gap in understanding how it operates during periods of global monetary tightening and uncertain timeframes. Most existing literature focuses on pre-2021 data, failing to capture the dynamics of the 2022–2024 global tightening cycle, this is the period where interest rates reached historic highs in both the US and Pakistan. Furthermore, while qualitative differences between developed and emerging markets are well-noted, there is a lack of comparative research that uses a unified econometric framework, specifically a Vector Error Correction Model (VECM) which aims to quantify the "speed of adjustment" and the long-run stability of these channels side-by-side.

1.3 Objectives of the Research

This research aims to bridge these gaps by pursuing the following objectives:

To empirically assess and understand the long-run equilibrium and short-run dynamics of the asset price channel in the US and Pakistan using monthly data from July 2015 to January 2025.

To compare the "speed of adjustment" (Error Correction Term) of the housing and equity markets in a developed economy versus an emerging one.

To analyze the permanence and magnitude of policy shocks through Impulse Response Functions (IRFs).

1.4 Significance of the Study

This study provides critical insights for policymakers at the State Bank of Pakistan (SBP) and the Federal Reserve, as it offers a data-driven comparison of how asset markets absorb policy shocks. For investors and financial analysts, the research quantifies the sensitivity of the PSX and Dow Jones to interest rate innovations, settled within a robust mathematical framework that accounts for long-term cointegration.

Chapter 2

Literature Review

The new asset price channel describes the procedure through which monetary policy affects real economic activity along the changes in asset valuations, investment, and consumption. Three major theoretical mechanisms can be distinguished: **Tobin's q theory, wealth effect** and **balance sheet channel**. The state of these mechanisms heavily depends on the financial market depth and institutional structure, implying far-reaching differences between developed and emerging economies.

Tobin's q theory relates monetary policy to investment through the definition of q: the ratio between a firm's market value and the replacement cost of capital (Tobin, 1969). In response, a decline in interest rates increases equity prices, which in turn raises q. This encourages firms to raise equity in order to finance new investments. The wealth effect works through household balance sheets: rising asset prices increase net wealth and consumption in a way that is consistent with the Life-Cycle Hypothesis (Modigliani and Brumberg, 1954). The balance sheet channel further strengthens transmission because of improved collateral values that reduce credit market frictions due to adverse selection and moral hazard (Bernanke and Gertler, 1989).

Evidence shows that the asset price channel is highly effective in the *United States of America*. Financial markets react swiftly to Monetary Policy Surprises, with an average stock price reduction of 1% from an unexpected rate hike of 25 basis points (Bernanke and Kuttner, 2005). Also, the housing wealth effect is highly significant because of the well-developed mortgage market in the US, implying an MPV of consumption in the range of USD 0.05 to 0.09 per additional unit of housing wealth (Federal Reserve, 2022;2024). After the Global Financial Crisis, research on Quantitative Easing depicts that extensive purchases of assets contributed largely to the enhancement of asset prices, thus implying the continuous relevance of the asset price channel, even when operating within the ZLB context, as asserted by (Gagnon et al.) in their research on QE conducted in 2011.

Pakistan's monetary transmission channel is, however, credit-driven with a weak asset price channel. This is supported by a research study done by the *State Bank of Pakistan*, concluding that changes in the policy rate affect stock prices marginally, given the fact that the Pakistan Stock Exchange is relatively smaller with minimal investor participation (SBP, 2022). Although real estate represents the biggest source of private wealth, the lack of a mortgage market impedes the realization of increases in wealth into consumption (Alam & Rashid, 2023). Furthermore, recent literature suggests that the spillover effects have become significant, with the decisions by the US Federal Reserve influencing Pakistan's financial markets more than the local Fed decisions, that is, the SBP (SBP, 2024).

Thus, it is revealed from the literature that while the asset price channel is recognized as playing a crucial role in managing aggregate demand in developed countries like the US, this channel in the case of Pakistan faces structural frictions. This is attributed to the lack of deep markets, the presence of a large informal economy, and the dominant role of bank credits in the economy. The result therefore highlights the significance of financial development for ensuring effective monetary transmission.

Chapter 3

Methodology

This section outlines the econometric framework and the specific statistical concepts and procedures used to investigate the Asset Price Monetary Transmission Mechanism in the United States and Pakistan. Given the non-stationary nature of macroeconomic time series, this study opts for a Vector Error Correction Model (VECM) to capture both the short-run dynamics and the long-run equilibrium relationships between monetary policy and asset valuations.

The dataset comprises a comprehensive set of macroeconomic variables designed to analyze the reaction between domestic monetary policy, global financial conditions, and real economic activity. The data for this research was extracted from four primary official databases namely, the State Bank of Pakistan (SBP) E-Data Portal for the KIBOR (6M) rates, the Pakistan Bureau of Statistics (PBS) for the LSMI, CPI (Base Year 2005), and both House Rent Indices (HRI 2000 and 2005). Moreover, United States of America data was extracted from the Federal Reserve Economic Data (FRED) for the US-based variables, including the Federal Funds Rate (FFR), Industrial Production Index (IPI), and US Inflation rate. Furthermore, Yahoo Finance for the historical price data of the PSX Index and the Dow Jones Industrial Average.

3.1. Model Specification:

The research is based on a 5 variable Vector Error Correction Model (VECM) system for each country to map the transmission of Monetary framework on asset price in both the nation. The variables include:

List of Variables

Pakistan's Model	US Model
KIBOR(6M)	FFR
PSX	Dow Jones
CPI	Inflation Rate
LSMI	IPI
HRI	HRI

Table 1

3.2. Econometric Procedure:

The procedure was conducted using several steps to ensure robustness and accuracy in the result.

1. Unit Root Test:
   To establish the order of integration, **Augmented Dickey-Fuller (ADF)** and **Phillips-Perron (PP)** tests are conducted. This step confirms that all series are integrated of the same order, specifically $I(1)$ assuring that they are non-stationary at levels but stationary at first difference.

2. Lag Length Selection:
   The optimal lag interval (1 to 4) is selected using the **Akaike Information Criterion (AIC)** and **Schwarz Criterion (SC)**.

3. Johansen Cointegration Test;
   The study uses the Johansen test in order to confirm the presence of a long-run relationship. Using the **Trace** and **Maximum Eigenvalue** statistics, the test evaluates the rank (r) of the cointegrating matrix. A result of $\boldsymbol{r = 1}$ is found that indicates a unique and stable economic bond between monetary policy and asset prices.

4. VECM Estimation and Innovation Analysis:
   Once above tests were conducted to ensure robustness, the VECM is estimated to derive the **Error Correction Terms**. The transmission mechanism is then visualized through:

- Impulse Response Functions (IRF): Tracing the response of PSX/Dow Jones and HRI to a one-standard-deviation shock in the policy rate.
- Variance Decomposition (FEVD): Quantifying the percentage of asset price volatility attributed to monetary policy shocks.

Chapter 4

Empirical Results and Discussion

This chapter goes through the empirical findings of the study which is structured to compare the transmission mechanisms of the United States and Pakistan. The analysis is followed by the sequential econometric steps outlined in the methodology chapter, focusing on stationarity, long-run equilibrium, and dynamic shock transmission.

4.1 Unit Root Test Results

In order to establish the order of integration for the time-series variables, both the Augmented Dickey-Fuller (ADF) and Phillips-Perron (PP) tests were conducted. The results for both the Pakistan and US datasets indicate that all variables—including policy rates (KIBOR/FFR), asset indices (PSX/DOWJ), and real estate proxies (HRI)—are non-stationary at levels.

However, upon taking the first difference, all variables become stationary with p-values of 0.0000, confirming they are integrated of order one ($I(1)$). This uniform stationarity across both economies provides a robust foundation for the subsequent cointegration analysis.

Results below show us that we were able to conclude that we can reject the null hypothesis of variables non-stationary behavior.

Unit Root Test Results

Variable	Country	Test Used	Prob. (at Level)	Prob. (at 1st Diff)	Conclusion
KIBOR / FFR	Pak / US	PP & ADF	>0.05	0.0001/0.0000	$I(1)$
PSX / DOWJ	Pak / US	PP & ADF	>0.05	0.0000 (both)	$I(1)$
HRI	Pak / US	PP & ADF	>0.05	0.0000 (both)	$I(1)$
CPI / Inflation	Pak / US	PP & ADF	>0.05	0.02/0.000	$I(1)$
LSMI / IPI	Pak / US	PP & ADF	>0.05	0.0000 (both)	$I(1)$

Table 2

4.2 Cointegration and Long-Run Relationship

The Johansen Cointegration test was applied to identify the existence of a stable long-run relationship.

- Pakistan: Both the Trace and Maximum Eigenvalue tests strongly reject the null hypothesis of no cointegration ($r = 0$) with p-values of 0.0064 and 0.0022, respectively. This unanimously confirms exactly one ($r = 1$) cointegrating equation, proving a stable long-run Asset Price Channel exists in Pakistan.
- United States: The Trace test confirms one cointegrating vector at the 0.05 level ($p = 0.0420$). While the Max-Eigen test shows weaker significance ($p = 0.1502$), the Trace result justifies the move to a VECM framework, indicating a "looser" but present long-run link in the highly efficient US market.

Johansen Cointegration Test

Country	Hypothesized No. of CE(s)	Trace Statistic	0.05 Critical Value	Prob.	Decision
Pakistan	None *	76.81	69.81	0.0064	Reject $H0$ $(r = 1)$
	At most 1	37.66	47.85	0.4053	Fail to Reject
USA	None *	70.76	69.81	0.0420	Reject $H0$ $(r = 1)$
	At most 1	41.19	47.85	0.1824	Fail to Reject

Table 3

4.3 VECM Estimation and Speed of Adjustment

The estimated **Vector Error Correction Model (VECM)** reveals the "speed of adjustment" back to equilibrium through the **Error Correction Term (ECT)**. This precisely tells us how quickly assets react to reach equilibrium. The below figures show us the VECM output.

Error Correction VECM Output

Error Correction	D(KIBOR/FFR)	D(PSX/DOWJ)	D(HRI)	Interpretation
Pakistan Model	-0.010	-55.80	-0.040	4% of housing error is fixed monthly.
(t-statistic)	[-1.23]	[-0.17]	[-3.30]	Result for HRI is highly significant.
USA Model	-0.011	60.61	-0.015	1.5% of housing error is fixed monthly.
(t-statistic)	[-2.00]	[2.18]	[-1.71]	Result for HRI is significant at 10%.

Table 4

4.4 Innovation Analysis: Impulse Response Functions (IRFs)

The Impulse Response Functions discussed below trace the response of endogenous variables to a one-standard-deviation shock in the policy rate (FFR for the US and KIBOR for Pakistan) over a 10-month horizon.

4.4.1 US Asset Price Channel

In the United States, the transmission mechanism exhibits high efficiency and immediate reaction times:

- Equity Market (DOWJ): The Dow Jones Industrial Average exhibits a sharp, immediate collapse following an FFR shock, plunging from approximately 400 to near 50 within the first two months. This immediate re-valuation confirms the presence of a highly efficient equity channel.

- Real Estate (HRI): The US House Rent Index shows a smooth, persistent, and steady decline throughout the 10-month period, indicating that monetary tightening has a predictable and cooling effect on the housing sector.
- Industrial Output (IPI) and Inflation: Inflation follows a classic downward trajectory, while Industrial Production (IPI) shows a short-term spike before settling, reflecting the initial shock to production costs.

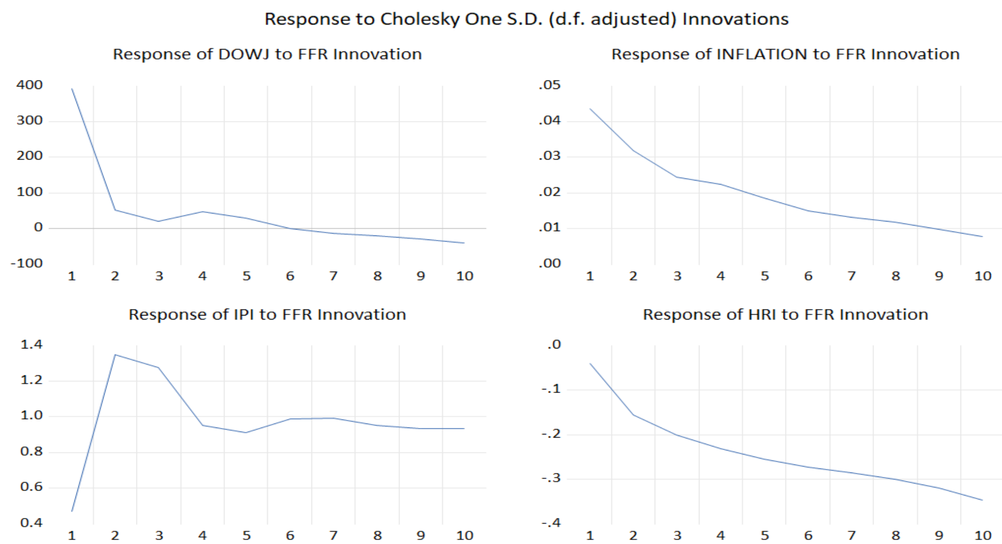

Figure 1 United States Impulse Responses to FFR innovation

4.4.2 Pakistan Asset Price Channel (Figures 5-8)

Contrary to the US, the Pakistan model below reveals significant structural lags and friction-heavy transmission:

- Equity Market (PSX): Unlike the immediate drop in the US, the Pakistan Stock Exchange (PSX) shows a gradual and lagged decline. The index continues to fall steadily through the entire 10-month horizon, suggesting that market participants in Pakistan incorporate policy changes over a longer duration.
- Real Estate (HRI): The response of the Pakistani House Rent Index is "jagged" and delayed as it initially spikes before falling downward, which confirms the "stickiness" identified in our earlier VECM speed of adjustment results.
- Price Puzzle (CPI): The most significant finding as the CPI in Pakistan shows a persistent upward trend following a KIBOR shock. This is common "Price Puzzle" phenomenon in emerging markets where higher interest rates increase the cost of doing business, which is then passed on to consumers.

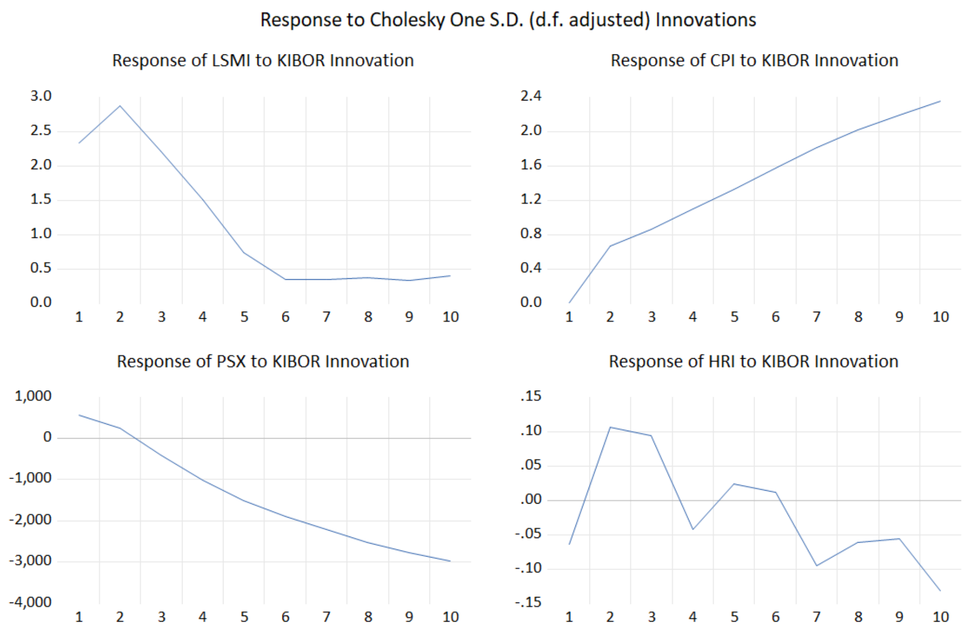

Figure 2 Pakistan's Impulse Responses to KIBOR innovation

Summary of Impulse Responses		
Variable	**US Response (FFR Shock)**	**Pakistan Response (KIBOR Shock)**
Equities	Immediate, sharp contraction.	Gradual, persistent decline.
Housing	Smooth, predictable cooling.	Delayed and jagged "sticky" response.
Prices	Classic downward trend.	Upward "Price Puzzle" effect.

Table 5

Chapter 5

Conclusion and Policy Recommendations

5.1 Conclusion

This study conducted a comparative empirical assessment of the Asset Price Monetary Transmission Mechanism in the United States and Pakistan from July 2015 to January 2025. Using a Vector Error Correction Model (VECM), the research identified a stable, long-run equilibrium relationship ($r = 1$) between monetary policy and asset valuations in both economies. However, the efficiency and speed of this transmission between both nations vary significantly.

The findings reveal that the US Asset Price Channel is highly fast and responsive, with equity markets (Dow Jones) depicting an immediate and significant contraction following a policy shock. In contrast, the Pakistan Stock Exchange (PSX) responds with significant lags, reflecting higher structural frictions. Furthermore, the real estate channel in Pakistan is notably "sticky," with a speed of adjustment of only 4% per month, compared to a more persistent but predictable cooling in the US housing market. Ultimately, while the transmission mechanism is functional in both nations, its potency in Pakistan is constrained by institutional maturity and market depth.

5.2 Policy Recommendations

For Pakistan:

- *Market Formalization:* The State Bank of Pakistan (SBP) should speed-up and strategize the documentation and formalization of the real estate sector. The "sticky" and jagged response of House Rent Index (HRI) explains that without a transparent, documented market, the wealth effect remains an unreliable tool for controlling aggregate demand.
- *Financial Depth:* To strengthen the equity channel's efficiency, policies should aim to increase retail participation in the PSX. The current lagged response to KIBOR innovations suggests that the market lacks the liquidity needed to price in policy changes as rapidly as developed markets.
- *Addressing the Price Puzzle:* Policymakers must take into consideration observed "Price Puzzle", this is where inflation initially rises after rate hikes by coordinating monetary tightening with supply-side interventions to manage the increased cost of business for local firms.

For the United States:

- *Managing Market Volatility:* Due to the extreme sensitivity of the Dow Jones to Federal Funds Rate (FFR) innovations, the Federal Reserve should continue with "Forward Guidance" to minimize sudden market collapses and ensure a more gradual re-valuation of assets.
- *Persistent Housing Cooling:* As the US HRI shows a very persistent long-term decline after a shock, the Fed must be cautious of "over-tightening," as the impact on the housing market continues to deepen well beyond the initial policy action.

5.3 Limitations of the Study

While the results are robust, certain statistical limitations must be noted. The diagnostic tests for both the US and Pakistan models indicated remaining **residual autocorrelation** and a **lack of normality** in the error terms. These issues are common in high-frequency monthly macroeconomic data, particularly in emerging markets where sudden political or external shocks create "fat tails" in the distribution. Additionally, the use of the **House Rent Index (HRI)** serves as a proxy for the broader real estate market, which may not fully capture the volatility of sale prices in undocumented sectors.

APPENDIX A

Time Frame	KIBOR (6M)	HRI (BY2005)	LSMI	PSX Index	CPI (BY2005)
UR(at first level)	0.0001	0.053	0	0	0.02
Jul-15	6.891	168.87	91	34432.55	1.90
Aug-15	7.03	168.87	96	35782.17	1.80
Sept-15	7.05	168.87	95	34713.76	1.30
Oct-15	6.8095	170.85	95.8	32287.41	1.60
Nov-15	6.5848	170.85	94.5	34303.52	2.70
Dec-15	6.463863	170.85	104	32263.56	3.20
Jan-16	6.521994	174.32	110	32841.49	3.30
Feb-16	6.424636	174.32	106	31417.13	4.00
Mar-16	6.356818	174.32	112	31384.67	3.90
Apr-16	6.356818	176.44	101	33129.75	4.20
May-16	6.361674	176.44	98.8	34727.2	3.20
Jun-16	6.307273	176.44	96.2	36046.97	3.20
Jul-16	6.122149	178.64	91.7	37887.84	4.10
Aug-16	6.039643	178.64	97	39567.45	3.60
Sept-16	6.025455	178.64	93.7	39813.6	3.90
Oct-16	6.053069	182.22	98.2	40652.88	4.20
Nov-16	6.08	182.22	100	39894.1	3.80
Dec-16	6.1395	182.22	108	43032.81	3.70
Jan-17	6.12	185.86	111	47881.53	3.70
Feb-17	6.12	185.86	113	48963.6	4.20
Mar-17	6.12	185.86	123	48627.9	4.90
Apr-17	6.15	188.68	107	48205.55	4.80
May-17	6.15	188.68	105	49146.38	5.00
Jun-17	6.15	188.68	102	50484.37	3.90
Jul-17	6.1445	191.49	106	46523.33	2.90
Aug-17	6.1505	191.49	108	46071.7	3.40
Sept-17	6.161579	191.49	103	41109.65	3.90
Oct-17	6.1682	194.05	108	42431.54	3.80
Nov-17	6.1764	194.05	102	39662.8	4.00
Dec-17	6.2058	194.05	114	40048.76	4.60
Jan-18	6.2336	195.96	125	40470.44	4.40
Feb-18	6.435789	195.96	118	44103.07	3.80
Mar-18	6.507143	195.96	127	43259.45	3.20
Apr-18	6.504762	201.96	118	45624.13	3.70
May-18	6.59381	201.96	109	45471.74	4.20
Jun-18	6.978421	201.96	101	42780.91	5.20
Jul-18	7.593	206	107	41948.67	5.80

Aug-18	8.046316	206	106	42742.89	5.80
Sept-18	8.229444	206	106	41731.76	5.10
Oct-18	9.433913	207.09	118	40985.03	6.80
Nov-18	9.87	207.09	109	41616.27	6.50
Dec-18	10.658	207.09	108	40383.27	6.20
Jan-19	10.76136	212.02	131	37067.98	7.27
Feb-19	10.85316	212.02	120	40810.7	8.83
Mar-19	10.89667	212.02	127	39083.98	11.17
Apr-19	11.248	214.78	122	38633.56	10.78
May-19	12.171	214.78	122	36776.22	10.91
Jun-19	13.063	214.78	108	35890.48	10.39
Jul-19	13.557	216.0465	109	33963.89	10.91
Aug-19	14.071	216.0465	106	31938.48	13.64
Sept-19	13.96	216.0465	103	29672.12	14.81
Oct-19	13.616	217.138247	110	32078.85	14.29
Nov-19	13.494	217.138247	105	34203.68	16.49
Dec-19	13.5	217.138247	114	39287.65	16.36
Jan-20	13.484	221.2152398	122	40735.08	18.96
Feb-20	13.485	221.2152398	121	41630.93	16.10
Mar-20	12.339	221.2152398	99.3	37983.62	13.25
Apr-20	9.4871	221.7952304	66.4	29231.63	11.04
May-20	7.9844	221.7952304	81.3	34111.64	10.65
Jun-20	7.77	221.7952304	95.4	33931.23	11.17
Jul-20	6.7914	225.3092911	107	34421.92	12.08
Aug-20	7.2025	225.3092911	95.8	39258.44	10.65
Sept-20	7.2873	225.3092911	107	41110.93	11.69
Oct-20	7.3357	227.2880826	111	40571.48	11.56
Nov-20	7.35	227.2880826	112	39888	10.78
Dec-20	7.348696	227.2880826	123	41068.82	10.39
Jan-21	7.3735	231.757422	126	43755.38	7.40
Feb-21	7.574211	231.757422	127	46385.54	11.30
Mar-21	7.785909	231.757422	122	45865.02	11.82
Apr-21	7.773333	235.561478	115	44587.85	14.42
May-21	7.675	235.561478	108	44262.35	14.16
Jun-21	7.695	235.561478	118	47896.34	12.60
Jul-21	7.628889	238.9390703	112	47356.02	10.91
Aug-21	7.57	238.9390703	114	47055.29	10.91
Sept-21	7.796364	238.9390703	115	47419.74	11.69
Oct-21	8.4655	242.3337212	117	44899.6	11.95
Nov-21	9.339091	242.3337212	119	46184.71	14.94
Dec-21	11.30783	242.3337212	130	45072.38	15.97
Jan-22	11.3115	245.9671917	138	44596.07	16.88
Feb-22	10.817	245.9671917	143	45374.68	15.84
Mar-22	11.88857	245.9671917	154	44461.01	16.49
Apr-22	13.61857	248.7136178	133	44928.83	17.40

May-22	14.888	248.7136178	129	45249.41	17.92
Jun-22	15.44429	248.7136178	132	43078.14	27.66
Jul-22	15.68235	252.3641469	110	41540.83	32.34
Aug-22	15.96333	252.3641469	114	40150.36	35.45
Sept-22	16.04714	252.3641469	112	42351.15	30.13
Oct-22	15.8119	255.0935145	116	41128.67	34.55
Nov-22	16.01	255.0935145	113	41264.66	30.91
Dec-22	17.00286	255.0935145	128	42348.63	31.82
Jan-23	17.3981	259.3240342	130	40420.45	35.84
Feb-23	18.7095	259.3240342	126	40673.06	40.91
Mar-23	21.2319	259.3240342	113	40510.37	45.97
Apr-23	22.10059	262.0022261	101	40000.83	47.27
May-23	22.08476	262.0022261	106	41580.85	49.35
Jun-23	23.03263	262.0022261	109	41388.2	38.18
Jul-23	23.07773	265.6015795	104	43683.78	36.75
Aug-23	23.6115	265.6015795	114	48326.51	35.58
Sept-23	22.49182	265.6015795	114	45005.74	40.78
Oct-23	21.54857	269.4909283	109	46294.62	34.81
Nov-23	21.5435	269.4909283	112	52075.16	37.92
Dec-23	20.98682	269.4909283	132	60667.83	38.57
Jan-24	21.4589	273.2267501	131	62696.99	36.75
Feb-24	21.4995	273.2267501	127	62054.25	30.00
Mar-24	21.62474	273.2267501	115	64755.31	26.88
Apr-24	21.39952	277.2013916	106	67162.7	22.47
May-24	21.09	277.2013916	114	71199.95	15.32
Jun-24	20.34706	277.2013916	108	76091.42	16.36
Jul-24	19.84	279.8966421	108	78531.93	14.42
Aug-24	18.59476	279.8966421	112	78101.16	12.47
Sept-24	16.9515	279.8966421	112	78715.16	8.96
Oct-24	14.50261	282.6430682	109	81199.43	9.35
Nov-24	13.39333	282.6430682	107	89292.74	6.36
Dec-24	12.33333	282.6430682	128	101921.2	5.32
Jan-25	11.85727	286.7371195	130	115258.1	3.12

Table 6 APPENDIX - PAKISTAN DATASET

APPENDIX B

Time Frame	Dow Jones	HRI (BY2000)	IPI (BY2017)	FFR	Inflation rate
UR(at first level)	0.00	0.00	0.0000	0.00	0.00
Jul-15	17689.86	174.479	101.2228	0.13	1.82
Aug-15	16528.03	174.918	101.0367	0.14	1.61
Sept-15	16284.7	175.023	100.7547	0.14	1.52
Oct-15	17663.54	175.027	100.2675	0.12	1.50
Nov-15	17719.92	175.116	99.5298	0.12	1.57
Dec-15	17425.03	175.089	99.0417	0.24	1.51
Jan-16	16466.3	175.013	99.5607	0.34	1.42
Feb-16	16516.5	175.261	99.0445	0.38	1.31
Mar-16	17685.09	176.584	98.3096	0.36	1.55
Apr-16	17773.64	178.48	98.6421	0.37	1.62
May-16	17787.2	180.329	98.4184	0.37	1.60
Jun-16	17929.99	181.901	98.8833	0.38	1.47
Jul-16	18432.24	183.003	98.9992	0.39	1.46
Aug-16	18400.88	183.641	98.8500	0.40	1.47
Sept-16	18308.15	183.928	98.7457	0.40	1.51
Oct-16	18142.42	183.997	98.7611	0.40	1.66
Nov-16	19123.58	184.206	98.3491	0.41	1.82
Dec-16	19762.6	184.379	99.0214	0.54	1.93
Jan-17	19864.09	184.633	98.7611	0.65	2.01
Feb-17	20812.24	185.001	98.3646	0.66	2.02
Mar-17	20663.22	186.51	99.0137	0.79	1.99
Apr-17	20940.51	188.524	100.0224	0.90	1.91
May-17	21008.65	190.519	100.1408	0.91	1.83
Jun-17	21349.63	192.243	100.3471	1.04	1.73
Jul-17	21891.12	193.487	100.1068	1.15	1.77
Aug-17	21948.1	194.32	99.6925	1.16	1.78
Sept-17	22405.09	194.79	99.7954	1.15	1.83
Oct-17	23377.24	195.055	101.0247	1.15	1.86
Nov-17	24272.35	195.419	101.2664	1.16	1.85
Dec-17	24719.22	195.816	101.4644	1.30	1.90
Jan-18	26149.39	196.088	101.4625	1.41	2.04
Feb-18	25029.2	196.879	101.7085	1.42	2.10
Mar-18	24103.11	198.545	102.2067	1.51	2.09
Apr-18	24163.15	200.584	103.3329	1.69	2.13
May-18	24415.84	202.42	102.4024	1.70	2.14
Jun-18	24271.41	204.013	103.1980	1.82	2.12
Jul-18	25415.19	204.922	103.3646	1.91	2.12
Aug-18	25964.82	205.301	104.0327	1.91	2.10
Sept-18	26458.31	205.347	104.1004	1.95	2.12

Oct-18	25115.76	205.342	103.9856	2.19	2.11
Nov-18	25538.46	205.079	104.0681	2.20	2.01
Dec-18	23327.46	204.669	104.0936	2.27	1.81
Jan-19	24999.67	204.17	103.4021	2.40	1.79
Feb-19	25916	204.395	102.8371	2.40	1.88
Mar-19	25928.68	205.743	102.8757	2.41	1.91
Apr-19	26592.91	207.658	102.2740	2.42	1.94
May-19	24815.04	209.321	102.3920	2.39	1.83
Jun-19	26599.96	210.555	102.4375	2.38	1.70
Jul-19	26864.27	211.304	101.9408	2.40	1.75
Aug-19	26403.28	211.662	102.6390	2.13	1.59
Sept-19	26916.83	211.842	102.2917	2.04	1.59
Oct-19	27046.23	211.937	101.4281	1.83	1.56
Nov-19	28051.41	212.078	101.9372	1.55	1.64
Dec-19	28538.44	212.21	101.7003	1.55	1.72
Jan-20	28256.03	212.368	101.0338	1.55	1.72
Feb-20	25409.36	213.187	101.3735	1.58	1.62
Mar-20	21917.16	215.163	97.4078	0.65	0.99
Apr-20	24345.72	217.208	84.5619	0.05	1.10
May-20	25383.11	218.456	85.9604	0.05	1.12
Jun-20	25812.88	219.782	91.5934	0.08	1.27
Jul-20	26428.32	221.537	95.0256	0.09	1.46
Aug-20	28430.05	224.016	95.9549	0.10	1.66
Sept-20	27781.7	226.767	95.9669	0.09	1.66
Oct-20	26501.6	229.776	96.7394	0.09	1.70
Nov-20	29638.64	232.279	97.0816	0.09	1.71
Dec-20	30606.48	234.328	98.3630	0.09	1.92
Jan-21	29982.62	236.4	98.8822	0.09	2.08
Feb-21	30932.37	239.18	95.6149	0.08	2.18
Mar-21	32981.55	244.178	98.3856	0.07	2.28
Apr-21	33874.85	249.787	98.5587	0.07	2.35
May-21	34529.45	255.412	99.4333	0.06	2.47
Jun-21	34502.51	261.132	99.7987	0.08	2.34
Jul-21	34935.47	265.463	100.2519	0.10	2.33
Aug-21	35360.73	268.733	100.0437	0.09	2.35
Sept-21	33843.92	271.381	98.8579	0.08	2.34
Oct-21	35819.56	273.587	100.1861	0.08	2.53
Nov-21	34483.72	275.954	100.8654	0.08	2.62
Dec-21	36338.3	278.527	100.5726	0.08	2.46
Jan-22	35131.86	281.904	100.1856	0.08	2.45
Feb-22	33892.6	287.123	100.8064	0.08	2.46
Mar-22	34678.35	294.918	101.3907	0.20	2.85
Apr-22	32977.21	301.537	101.4400	0.33	2.88
May-22	32990.12	306.299	101.3335	0.77	2.69
Jun-22	30775.43	308.095	101.0180	1.21	2.62

Jul-22	32845.13	306.932	101.2230	1.68	2.36
Aug-22	31510.43	303.51	101.0963	2.33	2.51
Sept-22	28725.51	300.346	101.2918	2.56	2.38
Oct-22	32732.95	298.565	101.2529	3.08	2.39
Nov-22	34589.77	296.751	100.9552	3.78	2.37
Dec-22	33147.25	294.244	99.7664	4.10	2.26
Jan-23	34086.04	292.739	100.5050	4.33	2.24
Feb-23	32656.7	293.536	100.6416	4.57	2.33
Mar-23	33274.15	297.459	101.0205	4.65	2.30
Apr-23	34098.16	301.573	101.2530	4.83	2.27
May-23	32908.27	305.439	100.9286	5.06	2.21
Jun-23	34407.6	308.398	100.1196	5.08	2.20
Jul-23	35559.53	310.32	100.9067	5.12	2.30
Aug-23	34721.91	311.683	100.8393	5.33	2.34
Sept-23	33507.5	312.538	101.0211	5.33	2.34
Oct-23	33052.87	312.889	100.4689	5.33	2.39
Nov-23	35950.89	312.125	100.8639	5.33	2.30
Dec-23	37689.54	310.939	100.6031	5.33	2.18
Jan-24	38150.3	310.809	99.2223	5.33	2.27
Feb-24	38996.39	312.736	100.2850	5.33	2.28
Mar-24	39807.37	316.919	100.4575	5.33	2.31
Apr-24	37815.92	320.85	100.2434	5.33	2.39
May-24	38686.32	323.794	100.8630	5.33	2.33
Jun-24	39118.86	325.347	100.8940	5.33	2.26
Jul-24	40842.79	325.668	99.9757	5.33	2.27
Aug-24	41563.08	325.111	100.4309	5.33	2.11
Sept-24	42330.15	324.743	99.8084	5.13	2.11
Oct-24	41763.46	324.061	99.4695	4.83	2.29
Nov-24	44910.65	323.777	99.2925	4.64	2.32
Dec-24	42544.22	323.375	100.3273	4.48	2.30
Jan-25	44544.66	323.691	100.0647	4.33	2.40

Table 7 APPENDIX B USA DATASET

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